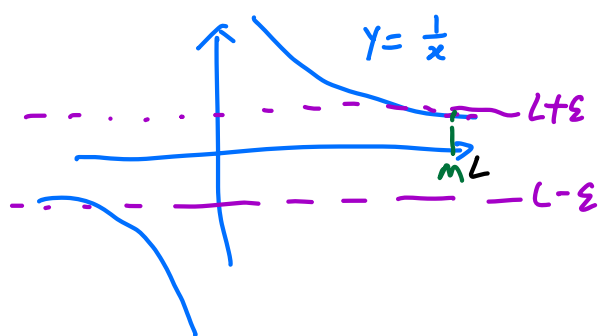


Lec 2-1 (Mon, Sep 8, 2025)

- Limits of infinity (2.6)
- Asymptotes: horizontal, vertical, oblique (2.6)
- Derivatives (3.1)

§2.6. Limits at infinity.



Intuition in symbol:

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0,$$

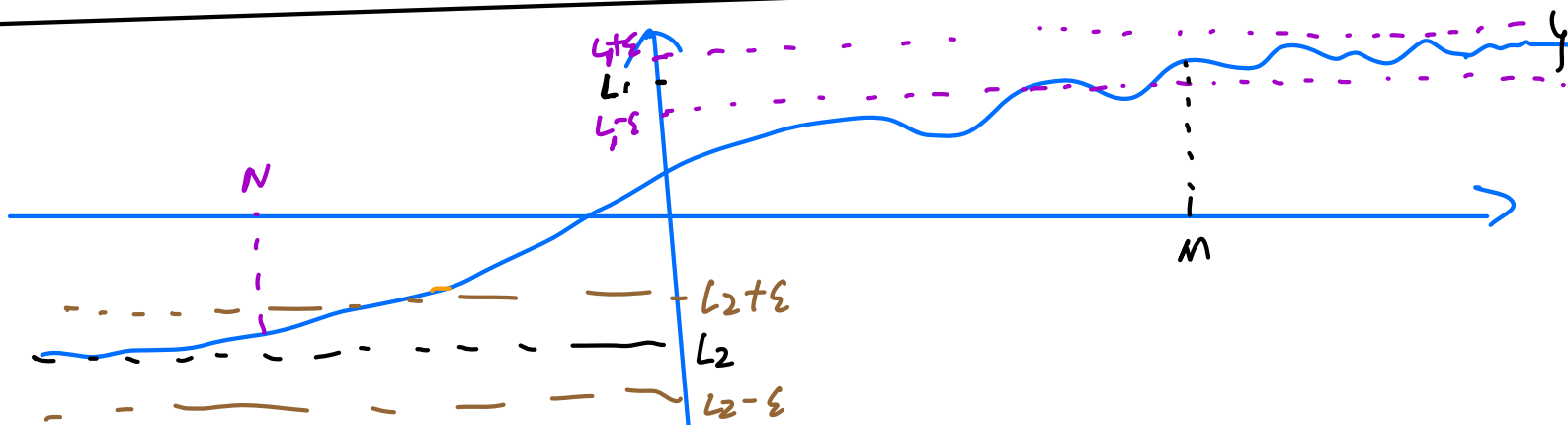
$$\lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$

Definition. ① We say that $f(x)$ has the limit L as x approaches infinity and write $\lim_{x \rightarrow \infty} f(x) = L$,

if for every number $\epsilon > 0$, there exists a corresponding number M , s.t.
 $|f(x) - L| < \epsilon$ for all $x > M$.

② We say that $f(x)$ has the limit L as x approaches minus infinity and write $\lim_{x \rightarrow -\infty} f(x) = L$,

if for every number $\epsilon > 0$, there exists a corresponding number N , s.t.
 $|f(x) - L| < \epsilon$ for all $x < N$.



• Prove that $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$ in ϵ - δ language. [Optional]

Pf: Let $\epsilon > 0$ be given. Set $M = \frac{1}{\epsilon}$.

For any $x > M$, we have

$$\left| \frac{1}{x} - 0 \right| = \frac{1}{x} < \frac{1}{M} = \varepsilon$$

Hence, $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$.

□

Similarly, we can formally show that $\lim_{x \rightarrow -\infty} \frac{1}{x} = 0$.

• Computation ?

— The limit laws also work for $\lim_{x \rightarrow \infty}$ or $\lim_{x \rightarrow -\infty}$

e.g.
$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow \infty} f(x)}{\lim_{x \rightarrow \infty} g(x)}$$

e.g.
$$\lim_{x \rightarrow \infty} \frac{1}{x^3} = \left(\lim_{x \rightarrow \infty} \frac{1}{x} \right)^3 = 0^3 = 0$$

— The Sandwich theorem also works for $\lim_{x \rightarrow \infty}$ or $\lim_{x \rightarrow -\infty}$.

e.g.
$$\lim_{x \rightarrow \infty} \frac{\sin x}{x} = ?$$

Solution: For all $x > 0$, we have $-\frac{1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$.

Then as $\lim_{x \rightarrow \infty} \frac{1}{x} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$, by sandwich theorem,

we have
$$\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$$

e.g.
$$\lim_{x \rightarrow \infty} x \cdot \left\lfloor \frac{1}{x} \right\rfloor = ?$$
 [Here $\lfloor x \rfloor$ = the biggest integer that is $\leq x$.

e.g. $\lfloor 4.99 \rfloor = 4$, $\lfloor 5 \rfloor = 5$, Floor function].

$$\lfloor -1.1 \rfloor = -2$$

Solution: we have $\frac{1}{x} - 1 < \left\lfloor \frac{1}{x} \right\rfloor \leq \frac{1}{x}$ for all $x \neq 0$.

Then for all $x > 0$, $x(\frac{1}{x} - 1) < x \cdot \lfloor \frac{1}{x} \rfloor \leq x \cdot \frac{1}{x}$

$$\text{As } \lim_{x \rightarrow 0^+} x(\frac{1}{x} - 1) = \lim_{x \rightarrow 0^+} (1 - x) = 1,$$

and $\lim_{x \rightarrow 0^+} x \cdot \frac{1}{x} = 1$, by sandwich theorem,

$$\text{we have } \lim_{x \rightarrow 0^+} x \cdot \lfloor \frac{1}{x} \rfloor = 1.$$

□

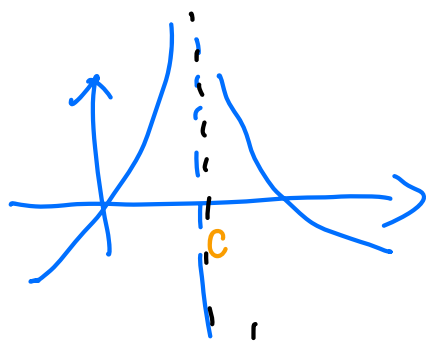
• Horizontal Asymptotes

Def: If $\lim_{x \rightarrow \infty} f(x) = b$ or $\lim_{x \rightarrow -\infty} f(x) = b$, then the line given by

$y = b$ is called a horizontal asymptote of $y = f(x)$.



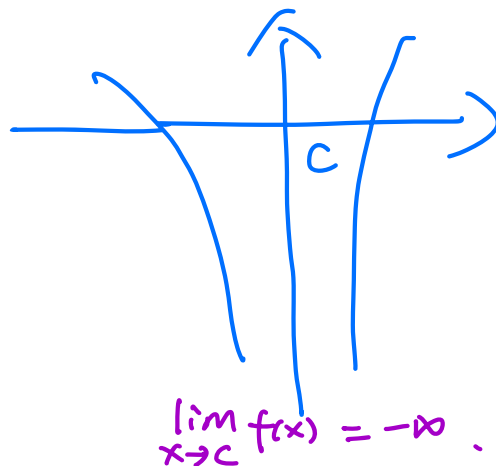
• Infinite limits; vertical asymptote.



$$\lim_{x \rightarrow c} f(x) = \infty.$$

$$\lim_{x \rightarrow c^+} f(x) = \infty$$

$$\lim_{x \rightarrow c^-} f(x) = -\infty$$



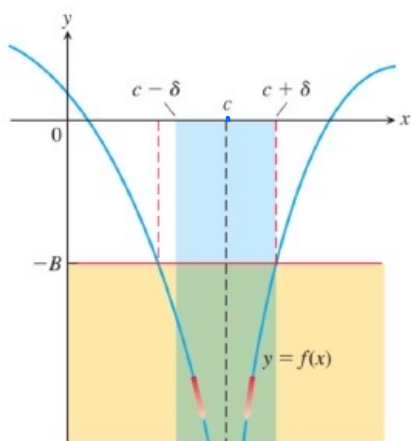


FIGURE 2.62 For $c - \delta < x < c + \delta$, the graph of $f(x)$ lies below the line $y = -B$.

DEFINITIONS

1. We say that $f(x)$ **approaches infinity** as x **approaches c** , and write

$$\lim_{x \rightarrow c} f(x) = \infty,$$

if for every positive real number B there exists a corresponding $\delta > 0$ such that for all x

$$0 < |x - c| < \delta \implies f(x) > B.$$

2. We say that $f(x)$ **approaches minus infinity** as x **approaches c** , and write

$$\lim_{x \rightarrow c} f(x) = -\infty,$$

if for every negative real number $-B$ there exists a corresponding $\delta > 0$ such that for all x

$$0 < |x - c| < \delta \implies f(x) < -B.$$

Definition. ① We say that $f(x)$ approaches infinity as x approaches c and write

$$\lim_{x \rightarrow c} f(x) = \infty$$

if for any positive real number B , there exists a corresponding $\delta > 0$, s.t.

$$f(x) > B \quad \text{for any } 0 < |x - c| < \delta \quad . \quad (\text{i.e. } x \in (c - \delta, c) \cup (c, c + \delta))$$

② We say that $f(x)$ approaches minus infinity as x approaches c and write

$$\lim_{x \rightarrow c} f(x) = -\infty$$

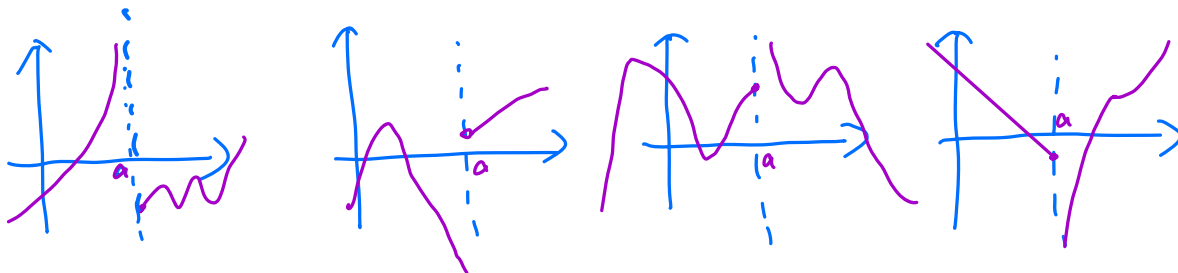
if for any negative real number $-B$, there exists a corresponding $\delta > 0$, s.t.

$$f(x) < -B \quad \text{for any } 0 < |x - c| < \delta \quad .$$

• Vertical asymptote

Definition. If $\lim_{x \rightarrow a^-} f(x) = \infty$ or $\lim_{x \rightarrow a^-} f(x) = -\infty$ or $\lim_{x \rightarrow a^+} f(x) = \infty$ or $\lim_{x \rightarrow a^+} f(x) = -\infty$, then the line $x=a$ is

called a vertical asymptote of $y=f(x)$.



For these four pictures, we can all say that

$x=a$ is a vertical asymptote of $y=f(x)$.

$\Rightarrow$ It is "OR" in the definition!

Summary. Horizontal asymptote : $\lim_{x \rightarrow \infty} f(x) = b$ or $\lim_{x \rightarrow -\infty} f(x) = b$.

vertical asymptote : $\lim_{x \rightarrow a^-} f(x) = \infty$ or $\lim_{x \rightarrow a^+} f(x) = -\infty$ or

$\lim_{x \rightarrow a^-} f(x) = -\infty$ or $\lim_{x \rightarrow a^+} f(x) = \infty$.

• Computation

It can be shown formally (using definitions) that the previous limit laws still hold when dealing with infinite limits.

Eg, If $\lim_{x \rightarrow c^+} f(x) = \infty$, $\lim_{x \rightarrow c^+} g(x) = \infty$, then

$$\lim_{x \rightarrow c^+} [f(x) + g(x)] = \lim_{x \rightarrow c^+} f(x) + \lim_{x \rightarrow c^+} g(x) = \infty.$$

We use the following **informal convention** (for convenience, not formal):

$$\cdot \quad \infty + \infty = \infty \quad ; \quad a \pm \infty = \pm \infty, \quad \forall a \in \mathbb{R}$$

$$\text{i.e.} \quad \begin{cases} a + \infty = \infty, & \forall a \in \mathbb{R} \\ a - \infty = -\infty, & \forall a \in \mathbb{R} \end{cases}$$

$$\cdot \quad a \times (\pm \infty) = \pm \infty, \quad \frac{\pm \infty}{a} = \pm \infty, \quad \forall a > 0$$

$$\cdot \quad a \times (\pm \infty) = \mp \infty, \quad \frac{\pm \infty}{a} = \mp \infty, \quad \forall a < 0.$$

$$\cdot \quad \infty \times (\pm \infty) = \pm \infty, \quad -\infty \times (\pm \infty) = \mp \infty.$$

$$\cdot \quad \frac{a}{\pm \infty} = 0, \quad \forall a \in \mathbb{R}$$

$$\cdot \quad \frac{a}{\pm 0} = \pm \infty, \quad \forall a > 0$$

if $\lim_{x \rightarrow c^+} f(x) = a^{>0}$, $\lim_{x \rightarrow c^+} g(x) = +0$, then

$$\lim_{x \rightarrow c^+} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c^+} f(x)}{\lim_{x \rightarrow c^+} g(x)} = \frac{a^{>0}}{+0} = \infty.$$

$$\cdot \quad \frac{a}{\pm 0} = \mp \infty, \quad \forall a < 0.$$

$$\cdot \quad \sqrt[n]{\infty} = \infty.$$

e.g.1. $\lim_{x \rightarrow -\infty} \frac{5x^2 + 8x - 3}{3x^2 + 2} =$

(if direct compute,

we have $\frac{\infty}{\infty}$, for

which we do not

know the result)

(we pick the highest order of the denominator: x^2)

limit laws

$$\lim_{x \rightarrow -\infty} \frac{5 + \frac{8}{x} - \frac{3}{x^2}}{3 + \frac{2}{x^2}}$$

$$\frac{5 + \lim_{x \rightarrow -\infty} \frac{8}{x} - \lim_{x \rightarrow -\infty} \frac{3}{x^2}}{3 + \lim_{x \rightarrow -\infty} \frac{2}{x^2}}$$

$$= \frac{5 + 0 - 0}{3 + 0}$$

$$= \frac{5}{3}$$

e.g.2. $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 1} - x) =$
 $\infty - \infty$

$$\lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + 1} - x)(\sqrt{x^2 + 1} + x)}{\sqrt{x^2 + 1} + x}$$

$$= \lim_{x \rightarrow \infty} \frac{\cancel{x^2 + 1} - \cancel{x^2}}{\sqrt{x^2 + 1} + x}$$

$$= 0$$

□

if we directly compute,

We have $\infty - \infty$.

again we do not

determine the result.

$$\lim_{x \rightarrow \infty} (2x - x) = \lim_{x \rightarrow \infty} x = \infty$$

$$\lim_{x \rightarrow \infty} (x - 3x) = \lim_{x \rightarrow \infty} (-2x) = -\infty$$

e.g.3 $\lim_{x \rightarrow 2} \frac{x-3}{x^2-4}$

$$= \frac{\lim_{x \rightarrow 2} (x-3)}{\lim_{x \rightarrow 2} (x^2-4)}$$

$$= \frac{-1}{\lim_{x \rightarrow 2} (x^2-4)}$$

$$= \frac{-1}{+0} = 0.001$$

$$-0 = -0.001$$

① $\lim_{x \rightarrow c} f(x)$ exists finitely

② $\lim_{x \rightarrow c} f(x) = \infty$

($\lim_{x \rightarrow c+} f(x) = (\lim_{x \rightarrow c-} f(x) = \infty)$)

we can directly compute this time.

Let us avoid using notation

" $\frac{-1}{0}$ " in the formal

proof.

e.g.4 $\lim_{x \rightarrow -\infty} \frac{2x^5+6x^4+1}{3x^2+x-7}$

③ $\lim_{x \rightarrow 2+} \frac{x-3}{x^2-4} = \frac{\lim_{x \rightarrow 2+} (x-3)}{\lim_{x \rightarrow 2+} (x^2-4)} \left(= \frac{-1}{0.01} = \frac{-1}{+0} \right) = -\infty$

$\lim_{x \rightarrow 2-} \frac{x-3}{x^2-4} = \frac{\lim_{x \rightarrow 2-} (x-3)}{\lim_{x \rightarrow 2-} (x^2-4)} \left(= \frac{-1}{-0.001} = \frac{-1}{-0} \right) = \infty$

$\therefore \lim_{x \rightarrow 2} \frac{x-3}{x^2-4}$ does not exist, even considering $\pm \infty$. (4)

$$= \lim_{x \rightarrow -\infty} \frac{2x^3+6x^2+\frac{1}{x^2}}{3+\frac{x}{x^2}-\frac{7}{x^2}}$$

$$= \frac{\lim_{x \rightarrow -\infty} (2x^3+6x^2) + \lim_{x \rightarrow -\infty} \frac{1}{x^2}}{\lim_{x \rightarrow -\infty} 3 + \lim_{x \rightarrow -\infty} \frac{1}{x} - \lim_{x \rightarrow -\infty} \frac{7}{x^2}}$$

$$= \frac{\lim_{x \rightarrow -\infty} 2x^2(x+3) + 0}{3 + 0 - 0}$$

$$= \frac{\lim_{x \rightarrow -\infty} 2x^2 \cdot \lim_{x \rightarrow -\infty} (x+3)}{3} \left(= \frac{\infty \cdot (-\infty)}{3} \right) = -\infty$$

(or:)

numerator: x^5

$$= \lim_{x \rightarrow -\infty} \frac{2+6 \cdot \frac{1}{x} + \frac{1}{x^5}}{3 \cdot \frac{1}{x^3} + \frac{1}{x^4} - \frac{7}{x^5}}$$

$$= \frac{2 + \lim_{x \rightarrow -\infty} 6 \cdot \frac{1}{x} + \lim_{x \rightarrow -\infty} \frac{1}{x^5}}{3 \lim_{x \rightarrow -\infty} \frac{1}{x^3} + \lim_{x \rightarrow -\infty} \frac{1}{x^4} - \lim_{x \rightarrow -\infty} \frac{7}{x^5}}$$

$$= \frac{2+0+0}{3 \lim_{x \rightarrow -\infty} \frac{1}{x^3} + \lim_{x \rightarrow -\infty} \frac{1}{x^4} - \lim_{x \rightarrow -\infty} \frac{7}{x^5}}$$

$$= \frac{2+0}{(0-)+0} = -\infty$$

Rmks :

1. The "rules" above are NOT real number operations.

They serve as a shortcut for you to remember and use the limit laws.

Therefore, try to NOT write " $\frac{1}{0}$ ", " $\frac{1}{\infty}$ ", etc in the formal proof. You can directly write down the answer from a last step.

the denominator is not a real "zero".
It means it has the "limit" zero.

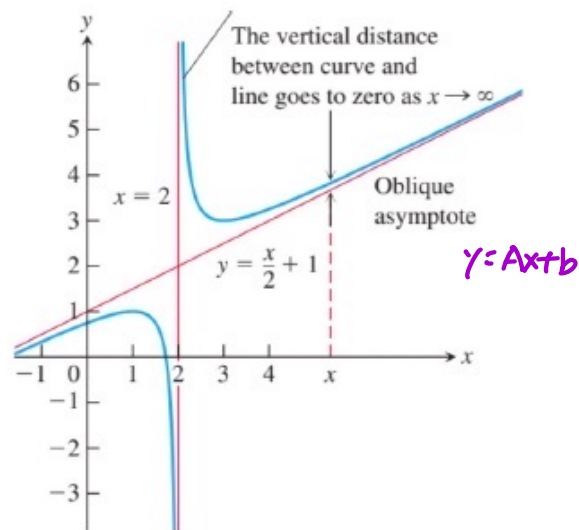
2. $\infty - \infty$, $0 \times \infty$, $\frac{0}{0}$, $\frac{\infty}{\infty}$ are "undefined"

(indeterminate forms : you do not determine the result and there are many possible answers. we will study this a few lectures later and will update our weapons !)

Oblique Asymptotes

More generally, an asymptote does not have to be horizontal or vertical. **oblique!**

$$\lim_{x \rightarrow \infty} [f(x) - Ax - B] = 0.$$



$$y = f(x) = \frac{x^2 - 3}{2x - 4}$$

Definition

The line given by $y = Ax + B$ is called an **oblique asymptote** of the graph of a function $y = f(x)$ if

$$\lim_{x \rightarrow \infty} [f(x) - Ax - B] = 0 \quad \text{or} \quad \lim_{x \rightarrow -\infty} [f(x) - Ax - B] = 0.$$

Q: Given f , how to compute an oblique asymptote?

Step 1. write down the definition

$$\lim_{x \rightarrow \infty} [f(x) - Ax - B] = 0.$$

Step 2. divide by x .

$$\lim_{x \rightarrow \infty} \frac{f(x) - Ax - B}{x} = \lim_{x \rightarrow \infty} \left(\frac{f(x)}{x} - A \right) - \lim_{x \rightarrow \infty} \frac{B}{x} = 0$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{f(x)}{x} - A = 0 \quad \Rightarrow \quad A \triangleq \lim_{x \rightarrow \infty} \frac{f(x)}{x}.$$

Step 3: compute B from definition.

$$B \triangleq \lim_{x \rightarrow \infty} [f(x) - Ax]$$

e.g. 5 Find the oblique asymptotes of the graph of

$$y = \frac{x^2 - 3}{2x - 4}.$$

Sol. Step 1: Let the oblique asymptote be $y = A_1 x + B_1$.

Then $\lim_{x \rightarrow \infty} \left[\frac{x^2 - 3}{2x - 4} - A_1 x - B_1 \right] = 0$

Step 2: divide by x .

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\frac{x^2 - 3}{2x - 4} - A_1 x - B_1}{x} &= \lim_{x \rightarrow \infty} \frac{x^2 - 3}{x(2x - 4)} - A_1 - \lim_{x \rightarrow \infty} \frac{B_1}{x} \\ &= \lim_{x \rightarrow \infty} \frac{1 - \frac{3}{x^2}}{2 - \frac{4}{x}} - A_1 - 0 = \frac{1}{2} - A_1 = 0 \end{aligned}$$

$$\therefore A_1 = \frac{1}{2}.$$

Step 3. Compute B from the definition.

$$B_1 = \lim_{x \rightarrow \infty} \left[\frac{x^2 - 3}{2x - 4} - \frac{1}{2} x \right]$$

$$= \lim_{x \rightarrow \infty} \frac{x^2 - 3 - x(x - 2)}{2x - 4}$$

$$= \lim_{x \rightarrow \infty} \frac{2x - 3}{2x - 4}$$

$$= \lim_{x \rightarrow \infty} \frac{2 - \frac{3}{x}}{2 - \frac{4}{x}} = 1.$$

$\therefore y = \frac{1}{2}x + 1$ is an oblique asymptote.

Similarly, assume $y = A_2x + B_2$ is an oblique asymptote as $x \rightarrow -\infty$. We have

$$\lim_{x \rightarrow -\infty} \left(\frac{x^2 - 3}{2x - 4} - A_2x - B_2 \right) = 0$$

$$\therefore \lim_{x \rightarrow -\infty} \frac{\frac{x^2 - 3}{2x - 4} - A_2x - B_2}{x} = 0 \Rightarrow A_2 = \frac{1}{2}.$$

$$\therefore B_2 = \lim_{x \rightarrow -\infty} \left(\frac{x^2 - 3}{2x - 4} - \frac{1}{2}x \right) = 1.$$

$\therefore y = A_2x + B_2 = \frac{1}{2}x + 1$ is also the oblique asymptote when $x \rightarrow -\infty$. 

$$\lim_{x \rightarrow \infty} (f(x) - Ax - B) = 0$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{f(x)}{x} - A - \frac{B}{x} \right) = 0.$$

When you are quite skilled, you can write

$$A_1 = \lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \frac{\frac{x^2-3}{2x-4}}{x} = \lim_{x \rightarrow \infty} \frac{x^2-3}{(2x-4)x} = \frac{1}{2}.$$

$$B_1 = \lim_{x \rightarrow \infty} (f(x) - A_1 x) = \lim_{x \rightarrow \infty} \left(\frac{x^2-3}{2x-4} - \frac{1}{2}x \right) = 1.$$

$$A_2 = \lim_{x \rightarrow -\infty} \frac{f(x)}{x} = \dots$$

$$B_2 = \lim_{x \rightarrow -\infty} (f(x) - A_1 x) = \dots$$

e.g. 6. Find the oblique asymptotes of the graph of

$$y = x + \sqrt{x}.$$

Sol: $A_1 = \lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \frac{x + \sqrt{x}}{x} = 1.$

$$B_1 = \lim_{x \rightarrow \infty} (f(x) - A_1 x) = \lim_{x \rightarrow \infty} (x + \sqrt{x} - 1x) = \lim_{x \rightarrow \infty} \sqrt{x} = \infty.$$

Hence, there is no oblique asymptote as $x \rightarrow \infty$.

As the function is not defined on $(-\infty, 0)$, there is also no oblique asymptote as $x \rightarrow -\infty$. $\square$

Summary for chap 2.

I) Motivation: rates of change / speeds.

Average speed = $\frac{f(b)-f(a)}{b-a}$ = the slope of the secant line of $y=f(x)$

Instant speed = $\lim_{\Delta t \rightarrow 0} \frac{f(t+\Delta t)-f(t)}{\Delta t}$ = the slope of the tangent line of $y=f(x)$

II) Theory: What is the limit, $\lim_{x \rightarrow c}$?

Limit

• finite $\lim_{x \rightarrow c} g(x) = L < \infty$

intuition: trend of a function

definition: ϵ - δ language (find a $\delta > 0$)

• one-sided $\lim_{x \rightarrow c^+} g(x)$

$\lim_{x \rightarrow c^-} g(x)$

Continuity

$\lim_{x \rightarrow c} g(x) = g(c) < \infty$

$\lim_{x \rightarrow c} g(x)$ exists and equals $g(c)$.

$\lim_{x \rightarrow c^+} g(x) = g(c)$

$\lim_{x \rightarrow c^-} g(x) = g(c)$

• limit at infinity

$$\lim_{x \rightarrow \infty} g(x)$$

or

$$\lim_{x \rightarrow -\infty} g(x)$$

[horizontal asymptote]

As $f(\infty)$ is not well defined and $f(c) < \infty$,

• infinite limit

$$\lim_{x \rightarrow c^+} g(x) = \infty$$

or

$$\lim_{x \rightarrow c^-} g(x) = \infty$$

or

$$\lim_{x \rightarrow c^+} g(x) = -\infty$$

or

$$\lim_{x \rightarrow c^-} g(x) = -\infty$$

[vertical asymptote]

We do not discuss the continuity of infinity at this moment

• intermediate value thm

$$\lim_{x \rightarrow \infty} [g(x) - Ax - B] = 0$$

[oblique asymptote]

• does not exist

$$\textcircled{1} \lim_{x \rightarrow c^-} g(x) \neq \lim_{x \rightarrow c^+} g(x)$$

$$\textcircled{2} \text{ e.g. } \lim_{x \rightarrow c^-} g(x) = -\infty$$

$$\lim_{x \rightarrow c^+} g(x) = \infty$$

$\textcircled{3}$ Oscillating

$$\text{e.g. } \lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$$

• discontinuity

$$\textcircled{1} \lim_{x \rightarrow c^-} g(x) = \lim_{x \rightarrow c^+} g(x) \neq g(c)$$

$$\textcircled{2} \lim_{x \rightarrow c^-} g(x) \neq \lim_{x \rightarrow c^+} g(x)$$

$\textcircled{3} \lim_{x \rightarrow c^-} g(x)$ or $\lim_{x \rightarrow c^+} g(x)$ does not exist as a finite number

$$\textcircled{1.1} \text{ e.g. } \lim_{x \rightarrow c^-} g(x) = -\infty$$

$$\textcircled{2.2} \text{ e.g. } \lim_{x \rightarrow 0^+} \sin\left(\frac{1}{x}\right) \text{ does not exist.}$$

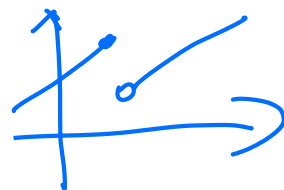
II) Computation

① direct substitution (continuous, non-zero denominator, ---)

② limit laws and composition of continuous outer function

$$g \text{ continuous} \Rightarrow \lim_{x \rightarrow c} g(h(x)) = g\left(\lim_{x \rightarrow c} h(x)\right)$$

③ "graph" observation (piecewise function)



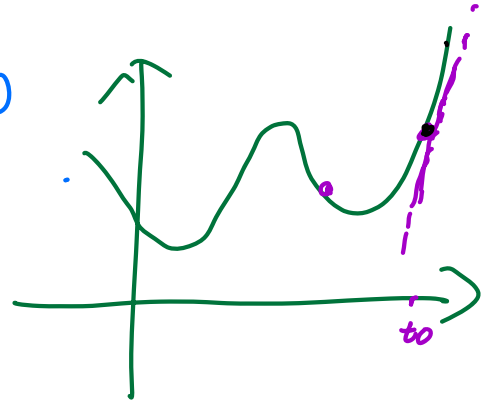
④ elimination of zero denominator ($\sqrt{x^2+1} - x$)

⑤ sandwich theorem $\left(\begin{array}{l} \text{e.g.} \\ \lim_{x \rightarrow c} f(x) = 0 \\ |g(x)| \leq m \end{array} \right) \Rightarrow \lim_{x \rightarrow c} f(x)g(x) = 0$

$$\textcircled{6} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

Chop 3.

Now we can study $\lim_{\Delta t \rightarrow 0} \frac{f(t_0 + \Delta t) - f(t_0)}{\Delta t}$



Derivatives

Def: Let $f : D \rightarrow \mathbb{R}$ be a function and let x_0 be an interior point of D . Then the derivative of f at x_0 , denoted by $f'(x_0)$, is defined by

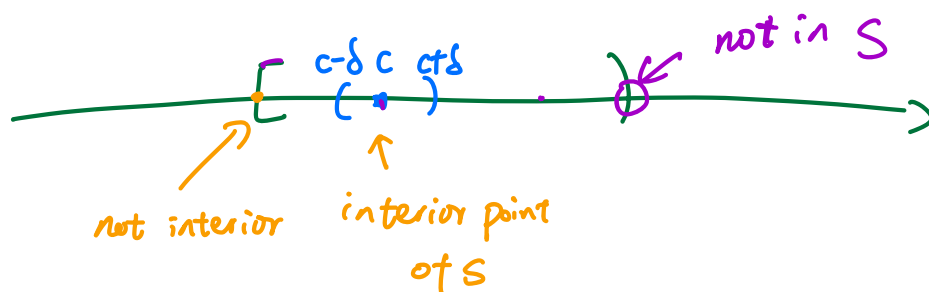
$$f'(x_0) \stackrel{\circ}{=} \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

[call it "f prime"]

$$= \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Def: Let $S \subseteq \mathbb{R}$. A point $c \in S$ is called an interior point of S if there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq S$.

Rmk. Every point in an open interval is an interior point.



As discussed in Lecture 1, for $y = f(x)$,

$f'(x_0)$ is • slope of the tangent line of the graph at $x = x_0$

• instant rate of change of y wrt x at $x = x_0$.

e.g. $f(x) = x^2$. What is $f'(1)$? $f'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 2$.

• Def (One-sided derivative)

Let x_0 be in the domain of f .

① The right-hand derivative of f at x_0 is denoted by $f'_+(x_0)$,

where

$$f'_+(x_0) \triangleq \lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h}$$

$$= \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0}$$

② The left-hand derivative of f at x_0 is denoted by $f'_-(x_0)$,

where

$$f'_-(x_0) \triangleq \lim_{h \rightarrow 0^-} \frac{f(x_0 + h) - f(x_0)}{h}$$

$$= \lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0}$$

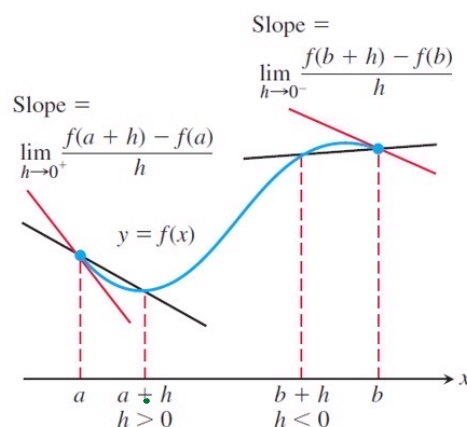
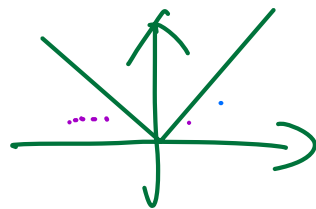


FIGURE 3.7 Derivatives at endpoints of a closed interval are one-sided limits.

e.g. 7. $f(x) = |x|$. What is $f'_+(0)$ and $f'_-(0)$?

Sol': $f'_+(0) = \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x - 0}{x - 0} = 1.$

$$f'_-(0) = \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{-x - 0}{x - 0} = -1$$



As $f'_+(0) \neq f'_-(0)$, $f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist. $\square$

e.g. 8. $f(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$. What is $f'(0)$?

Sol': $f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0}$

$$= \lim_{x \rightarrow 0} \frac{x^2 \sin(\frac{1}{x}) - 0}{x - 0}$$
$$= \lim_{x \rightarrow 0} x \underbrace{\sin(\frac{1}{x})}_{\text{bdd by } [-1, 1]}$$
$$= 0$$

e.g. 8. $f'(0)$ exists (f continuous at 0).

e.g. 9. $f'(0)$ does NOT exist (f cont at 0).

e.g. 8.5 $f(x) = \begin{cases} \sin \frac{1}{x} & , x \neq 0 \\ 0 & , x = 0 \end{cases}$ (f is NOT cont at 0)

e.g.9. $f(x) = \begin{cases} x \sin \frac{1}{x} & , \text{ if } x \neq 0 \\ 0 & , \text{ if } x = 0. \end{cases}$

What is $f'(0)$?

Sol': $f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{x \sin \frac{1}{x} - 0}{x - 0} = \lim_{x \rightarrow 0} \sin \frac{1}{x}$

which does not exist for both left-hand and right-hand sides. $\square$

[Note that f is continuous at 0] $\therefore \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0 = f(0)$.

e.g.10. $f(x) = \sqrt{x}$. What is $f'_+(0)$?

Sol': $f'_+(0) = \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{x} \quad \lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x}} = \infty$,

$\therefore f'_+(0)$ does not exist as a finite number. $\square$

Def: Let f be a function.

- If $f'(x_0)$ exists as a finite number, then f is said to be differentiable at x_0 .
- If $f'_+(x_0)$ exists as a finite number, then f is said to be right-differentiable at x_0 .
- If $f'_-(x_0)$ exists as a finite number, then f is said to be left-differentiable at x_0 .
- f is said to be differentiable on D if f is differentiable at every interior point of D and f is one-sided differentiable at every endpoint.

[Open intervals have no endpoint.]